# Low-Level Design (LLD)

Configuration, rules, workflows and runbook

| | |
|---|---|
| Document | Low-Level Design (LLD) |
| Project | SailPoint IdentityIQ - HR CSV authoritative source, lifecycle and LDAP ApacheDS provisioning |
| Version | 1.1 |
| Date | 23-10-2026 |
| Author | Ashok Kumar |
| Source | github.com/ashok8033/Study - branch main, folder SailPointIIQ-HR-CSV |

# 1. Environment

| Item | Value |
|---|---|
| IIQ URL | http://localhost:8080/identityiq (Tomcat on Windows) |
| Source package | SailPointIIQ-HR-CSV/ (config/, data/, ldif/) |
| Master import file | config/init-HR-CSV.xml (run from the IIQ console: import .../config/init-HR-CSV.xml) |
| HR file | C:/SailPoint/hrfeed/HR_Employees.csv |
| ApacheDS | ldap://localhost:10389, bind uid=admin,ou=system / secret (lab default) |
| Logs | WEB-INF/classes/log4j2.properties -> C:/SailPoint/logs/sailpoint.log |

# 2. HR Authoritative CSV application

| Setting | Value |
|---|---|
| Connector | sailpoint.connector.DelimitedFileConnector (type DelimitedFile), authoritative |
| file / delimiter / hasHeader | C:/SailPoint/hrfeed/HR_Employees.csv / \| / true |
| Account ID / display | employeeNumber |
| Rules | Correlation, Creation (IdentityCreation), Customization, Manager Correlation |
| Provisioning policy | Create Account Policy (fields from identity; date validation DD-MM-YYYY) |

**File layout**

```
employeeNumber|firstName|lastName|startDate|endDate|managerId|jobDescription|country|employeeType|isManager
1000|Ashokkumar|Chidambaram|01-01-2014|||Managing Director|India|Employee|Y
1001|Rajesh|Kumar|01-04-2015||1000|Chief Executive Officer|India|Employee|N
```

| Schema attribute | Notes |
|---|---|
| employeeNumber, firstName, lastName, startDate, endDate, managerId, jobDescription, country, employeeType, isManager | From the file |
| status, inactive, IIQDisabled | Calculated by HR CSV - Customization (Terminated -> IIQDisabled = true) |

# 3. Identity attributes (ObjectConfig Identity)

Created by the setup rule **HR CSV - Setup Identity Mappings** (run once; safe to re-run). They are shown on View Identity by **HR CSV - Show Attributes On Identity Page** (UIConfig identityViewAttributes).

| Identity attribute | Source | Target |
|---|---|---|
| employeeNumber, startDate, endDate, jobDescription, country, employeeType, isManager | HR CSV attribute of the same name | - |
| firstname / lastname / manager | HR firstName / lastName / managerId | - |
| inactive | Global rule Emergency Identity - Inactive Attribute (true if emergencyDisable, else HR inactive) | - |
| emergencyDisable | No source; editable (Permanent); set by the emergency workflow | - |
| displayName | Global rule HR CSV - Display Name | LDAP ApacheDS: displayName |
| status | Global rule HR Lifecycle - Status Attribute | - |
| email | LDAP ApacheDS: mail (generated by the Create Account policy) | - |

**Status calculation** (whole days; still Active on the end date):

```
if endDate  < today  -> Terminated
if startDate > today -> PreHire
otherwise            -> Active
```

# 4. Rules

| Rule | Type | Purpose |
|---|---|---|
| HR CSV - Customization | ResourceObjectCustomization | Trim values, validate dates, normalise isManager Y/N, set status / inactive / IIQDisabled |
| HR CSV - Correlation | Correlation | identityName = employeeNumber |
| HR CSV - Manager Correlation | ManagerCorrelation | Manager identity name = managerId |
| HR CSV - Identity Creation | IdentityCreation | Name = employeeNumber, first/last/display name, type |
| HR Lifecycle - Status Attribute | IdentityAttribute (global) | PreHire / Active / Terminated from HR dates |
| HR CSV - Display Name | IdentityAttribute (global) | firstName + ' ' + lastName |
| LDAP ApacheDS - Correlation | Correlation | identityName = LDAP employeeNumber |
| LDAP ApacheDS - Library | Library | JNDI connect, group add/remove, disable/enable, read attributes, time format |
| LDAP ApacheDS - Before Provisioning | BeforeProvisioning | Remove 'groups' from the request; keep changes as hrGroupsAdd / hrGroupsRemove |
| LDAP ApacheDS - After Provisioning | AfterProvisioning | Apply group changes as uniqueMember add/remove |
| LDAP ApacheDS - Customization | ResourceObjectCustomization | Read groups and account status (pwdAccountLockedTime etc.) |
| LDAP ApacheDS - Set Account Status | Called rule | Disable / enable LDAP account and update the IIQ link |
| LDAP ApacheDS - Disable Terminated Users | Utility | Catch-up: disable LDAP for identities already Terminated |
| HR CSV - Setup Identity Mappings | Setup | Create identity attributes, sources, global rules and LDAP target |
| HR CSV - Show Attributes On Identity Page | Setup | Add attributes to UIConfig identityViewAttributes |
| HR CSV - Diagnose | Utility (read-only) | Mappings (DB / cache), HR link, identity values, status rule result |
| Aggregate Single Account | Utility | Targeted aggregation of one account for any application |
| HR Lifecycle - Send Test Email | Utility | Show mail settings and send one test email |
| Emergency Identity - Inactive Attribute | IdentityAttribute (global) | inactive = emergencyDisable OR HR inactive |
| Emergency Identity - Provision Accounts | Called rule | Disable / enable every non-authoritative account; returns status + per-account details |
| Emergency Identity - Audit | Called rule | Write the AuditEvent and the log line |
| Emergency Identity - Setup | Setup | Create emergencyDisable, switch inactive to the global rule, add audit actions |

# 5. LDAP ApacheDS application

| Setting | Value |
|---|---|
| Connector | sailpoint.connector.LDAPConnector (type LDAP) |
| host / port / SSL | localhost / 10389 / false; authorizationType simple |
| Account search DN / filter | ou=people,dc=example,dc=com / (objectClass=inetOrgPerson), SUBTREE |
| Group search DN / filter | ou=groups,dc=example,dc=com / (objectClass=groupOfUniqueNames); member attribute uniqueMember |
| beforeProvisioningRule / afterProvisioningRule | LDAP ApacheDS - Before Provisioning / After Provisioning (application attributes) |
| CustomizationRule / CorrelationRule | LDAP ApacheDS - Customization / Correlation |

**Account schema** (identityAttribute dn, display uid): uid, cn, sn, givenName, displayName, mail, employeeNumber, employeeType, title, objectClass (multi), groups (entitlement, managed, multi), accountStatus, pwdAccountLockedTime, lockedTime, lastFailedLogin, passwordLastChanged, createdOn, modifiedOn.

**Create Account Policy**

| Field | Value |
|---|---|
| uid | firstname.lastname (lower case), made unique with 2, 3, ... against existing LDAP links |
| dn | uid=<uid>,ou=people,dc=example,dc=com |
| objectClass | top, person, organizationalPerson, inetOrgPerson |
| cn / displayName / sn / givenName | identity display name / display name / last name / first name |
| mail / employeeNumber / title / employeeType | <uid>@example.com / identity name / jobDescription / employeeType |
| password | Welcome@<employeeNumber> (lab only) |

**Account status mapping**

| pwdAccountLockedTime | accountStatus | IIQ flags |
|---|---|---|
| not present | Active | - |
| 000001010000Z (admin lock) | Disabled | IIQDisabled = true |
| any other timestamp (failed logins) | Locked | IIQLocked = true; lockedTime set |

# 6. ApacheDS directory layout (LDIF)

```
dc=example,dc=com
  ou=people   users created by IIQ: uid=<first.last>
  ou=groups   groupOfUniqueNames (placeholder member uid=admin,ou=system)
              cn=AllEmployees, cn=Contractors, cn=Managers, cn=Office365Access
```

Files: ldif/ApacheDS-ou-people-and-groups.ldif, ldif/ApacheDS-group-Office365Access.ldif. The dc=example,dc=com partition must exist (Studio: server configuration > Partitions).

# 7. Roles and entitlements

| Role | Type | Assignment / requirement | Entitlement |
|---|---|---|---|
| Birthright - All Active Users | business | MatchTerm status = Active | requires LDAP - All Employees |
| Birthright - Contractors | business | status = Active AND employeeType = Contractor | requires LDAP - Contractors |
| Birthright - Managers | business | status = Active AND isManager = Y | requires LDAP - Managers |
| Office365 Access | business | No selector - requestable only | requires LDAP - Office365Access |
| LDAP - All Employees / Contractors / Managers / Office365Access | it | Profile on LDAP ApacheDS | groups CONTAINS_ALL cn=...,ou=groups,dc=example,dc=com |

ManagedAttribute: groups = cn=Office365Access,ou=groups,dc=example,dc=com (requestable, with description).

# 8. Lifecycle event, workflow and email

| Object | Configuration |
|---|---|
| IdentityTrigger 'HR Lifecycle - Status Change' | type AttributeChange, attributeName status, handler WorkflowTriggerHandler, HandlerParameters/Attributes: workflow = HR Lifecycle - Status Change Notification |
| Workflow 'HR Lifecycle - Status Change Notification' | Collect Details -> [Disable LDAP \| Enable LDAP] -> Check Notify -> Send Email -> Stop |
| EmailTemplate 'HR Lifecycle - Status Change' | Name, employee number, old / new status, dates, job, manager, LDAP result |
| Recipient | notifyEmail = ashokhearts@gmail.com |

```
Collect Details : oldStatus / newStatus from the IdentityChangeEvent
Disable LDAP    : newStatus = Terminated and oldStatus != Terminated
                  -> rule LDAP ApacheDS - Set Account Status (action Disable)
Enable LDAP     : oldStatus = Terminated and newStatus = Active
                  and emergencyDisable is not true
                  -> rule LDAP ApacheDS - Set Account Status (action Enable)
Check Notify    : email only when oldStatus and newStatus are set and differ
```

# 9. Emergency Identity Disable / Enable

| Object | Configuration |
|---|---|
| SPRight / Capability | EmergencyIdentityDisableEnable / EmergencyIdentityAdministrator |
| DynamicScope 'Emergency Identity Administrators' | capabilities.name = SystemAdministrator OR EmergencyIdentityAdministrator |
| QuickLink 'Emergency Identity Disable / Enable' | action workflow, category Manage, workflowName = Emergency Identity Status Management Workflow |
| Workflow 'Emergency Identity Status Management Workflow' | transient; no approvals; form owner = launcher |
| File | config/QuickLink/QuickLink-Emergency_Identity-DisableEnable.xml, config/Workflow/Workflow-Emergency_Identity-StatusManagement.xml |

```
Request Form      : targetIdentity (Identity, postBack), read-only Identity Name, Current Status,
                    Emergency Disable, HR Status; requestedAction (only the valid value);
                    justification (required). Cancel -> Stop
Validate Request  : launcher has SystemAdministrator / EmergencyIdentityAdministrator; target exists,
                    is not the launcher; action matches emergencyDisable -> else DENIED -> Audit
Update Identity   : ObjectUtil.lockIdentity; emergencyDisable = (action == Disable);
                    inactive = rule Emergency Identity - Inactive Attribute; unlock (commits)
Provision Accounts: rule Emergency Identity - Provision Accounts -> provisioningResult
Collect Result    : workflowResult = SUCCESS | PARTIAL | FAILED | NOTHING_TO_DO
Audit             : rule Emergency Identity - Audit
Show Result       : read-only form for the requestor -> Stop
```

**Account handling** (Emergency Identity - Provision Accounts)

| Account | Disable | Enable |
|---|---|---|
| Authoritative (HR Authoritative CSV) | skipped | skipped |
| Already in the requested state | skipped | skipped |
| Any account, HR status Terminated | processed | skipped (stays disabled) |
| LDAP ApacheDS | rule LDAP ApacheDS - Set Account Status (Disable) | same rule (Enable) |
| Connector supports ENABLE | AccountRequest Disable via Provisioner | AccountRequest Enable via Provisioner |
| Other | MANUAL (counted as failed) | MANUAL (counted as failed) |

**Audit event**

| Field | Value |
|---|---|
| action | EmergencyIdentityDisable / EmergencyIdentityEnable / EmergencyIdentityDenied |
| source / target / created | requestor / target identity / date and time |
| string1 | workflow result: SUCCESS, PARTIAL, FAILED, NOTHING_TO_DO, DENIED |
| string2 / string3 / string4 | justification / emergencyDisable before -> after / account summary |
| attributes | requestedAction, accounts (full list), workflowCaseId |

# 10. Tasks

| Task | Type | Key options |
|---|---|---|
| HR Authoritative CSV - Account Aggregation | AccountAggregation | applications = HR Authoritative CSV, checkDeleted, noOptimizeReaggregation |
| HR Lifecycle - Daily Identity Refresh | Identity refresh | promoteAttributes, processTriggers, correlateEntitlements, provision, synchronizeAttributes, refreshManagerStatus |
| LDAP ApacheDS - Account Aggregation | AccountAggregation | applications = LDAP ApacheDS, checkDeleted |

**Recommended schedule:** HR aggregation -> Daily Identity Refresh (00:05) -> LDAP aggregation.

# 11. Deployment runbook

| Step | Action |
|---|---|
| 1 | ApacheDS: create the dc=example,dc=com partition; import both LDIF files; enable Password Policy |
| 2 | Copy data/HR_Employees.csv to C:/SailPoint/hrfeed/ |
| 3 | IIQ console: import .../config/init-HR-CSV.xml |
| 4 | Applications > LDAP ApacheDS: re-enter bind password, Test Connection; check the Rules tab |
| 5 | rule "HR CSV - Setup Identity Mappings" then rule "HR CSV - Show Attributes On Identity Page"; log out / in |
| 6 | Global Settings > Mail Settings: smtp.gmail.com, 587, TLS, app password; rule "HR Lifecycle - Send Test Email" |
| 7 | Run HR aggregation, Daily Identity Refresh, LDAP aggregation |
| 8 | rule "LDAP ApacheDS - Disable Terminated Users" once for existing leavers |
| 9 | Schedule the three tasks |
| 10 | rule "Emergency Identity - Setup" (after step 5); give non-admin operators the Emergency Identity Administrator capability |
| 11 | Run the Daily Identity Refresh once so inactive is recalculated by the new global rule |

# 12. Test cases

| # | Scenario | Steps | Expected result |
|---|---|---|---|
| T1 | Initial load | HR aggregation + refresh | 16 identities named 1000-1015; Ashok is everyone's manager |
| T2 | Status | Diagnose rule for 1000 | status = Active; 1009/1011/1015 Terminated; 1014 PreHire |
| T3 | Joiner | Set 1014 startDate to today; aggregate + refresh | Status Active; LDAP uid=james.taylor created in cn=AllEmployees; email |
| T4 | Leaver | Set endDate to yesterday; aggregate + refresh | Status Terminated; HR account disabled; LDAP pwdAccountLockedTime = 000001010000Z; groups removed; email |
| T5 | Rehire | Clear endDate; aggregate + refresh | Status Active; LDAP lock removed; email |
| T6 | Name change | Change lastName; aggregate + refresh | Identity last / display name updated; LDAP displayName updated |
| T7 | Request | Request Office365 Access; approve | User in cn=Office365Access |
| T8 | Status read-back | Lock an entry in Studio; LDAP aggregation | accountStatus Disabled / Locked in IIQ |
| T9 | Emergency access | Log in as a user without the capabilities | Quick link not shown |
| T10 | Emergency disable | spadmin: quick link, active user, Disable | Only Disable offered; emergencyDisable = true; inactive; LDAP locked; SUCCESS; audit EmergencyIdentityDisable |
| T11 | Emergency enable | Same user again, Enable | Only Enable offered; emergencyDisable = false; LDAP unlocked; audit EmergencyIdentityEnable |
| T12 | Refresh keeps state | Emergency-disable a user; run Daily Identity Refresh | Identity stays inactive; LDAP stays locked |

# 13. Troubleshooting

| Symptom | Cause / fix |
|---|---|
| Import: 'must match DOCTYPE root null' | File has no DOCTYPE - every file in the package now has one |
| Unable to open filestream | File path wrong or under OneDrive; use C:/SailPoint/hrfeed and check with Test-Path |
| could not resolve property: employeeNumber | Non-searchable attribute; correlation now uses identity name |
| Attributes missing on identity | Run the setup rule, then the Daily Identity Refresh; check with HR CSV - Diagnose |
| LDAP NO_SUCH_ATTRIBUTE groups | Handled by Before / After Provisioning rules |
| Account not updated after HR change | IIQ reads a different file; check File Path, schema Preview, Aggregate Single Account output |
| pwdAccountLockedTime not visible in Studio | Enable 'Fetch operational attributes' in the connection's Browser Options |
| No email | Mail settings / Gmail app password; run HR Lifecycle - Send Test Email |
| Emergency quick link not shown | User lacks SystemAdministrator / EmergencyIdentityAdministrator; log out and in after granting |
| Emergency result PARTIAL / MANUAL | An application has no ENABLE feature or provisioning failed; see the result lines and the audit event |
| Refresh re-activates an emergency-disabled identity | inactive still mapped from HR; run rule "Emergency Identity - Setup" |
| Java heap space | setenv.bat CATALINA_OPTS -Xms1024m -Xmx4096m, or tomcat9w.exe Java tab for the service |